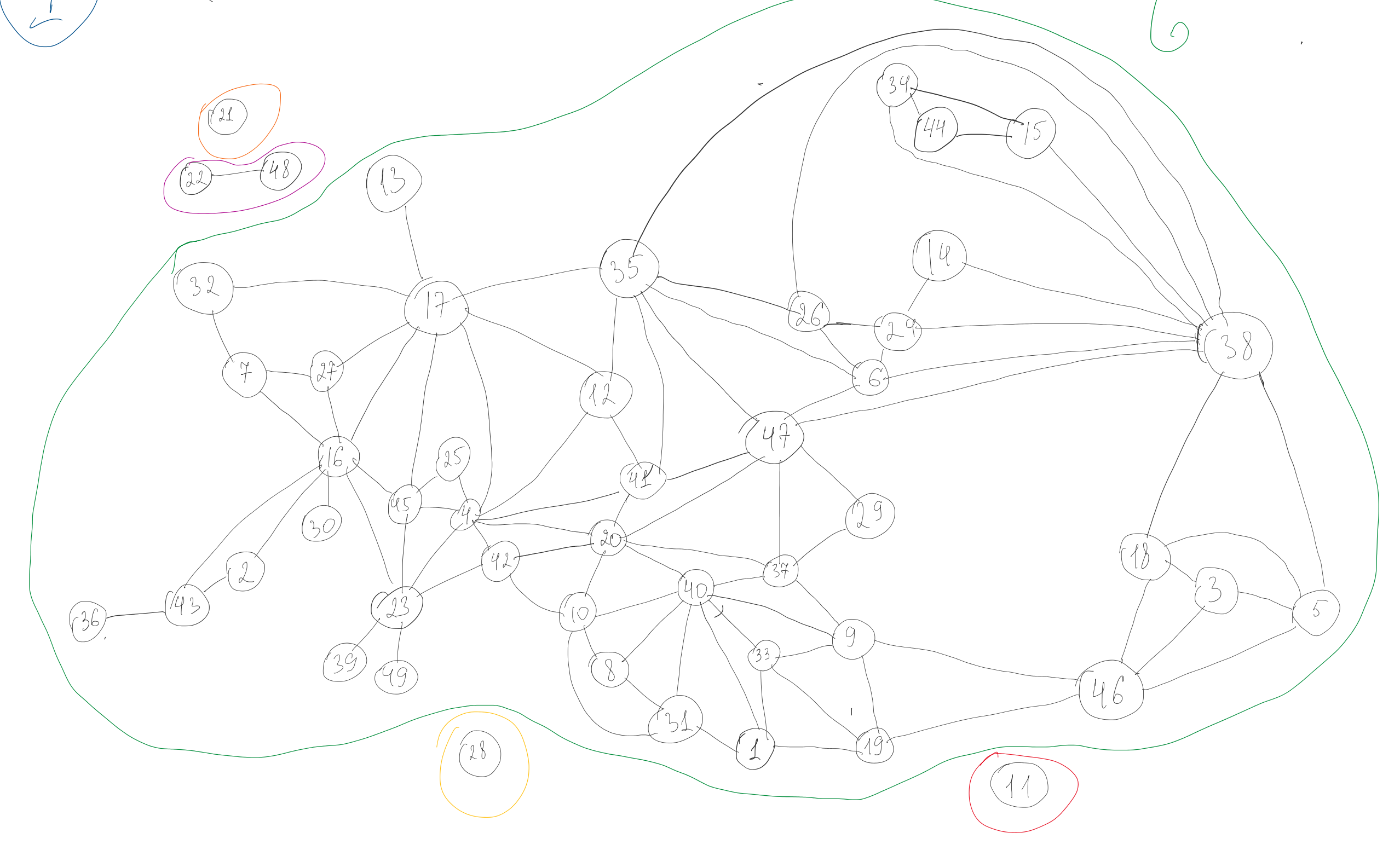
## I

(таблица для построения графа внизу, после работы)



## II

$|V| = 44$, $|E| = 91$ } (для G)

$|V| = 49$, $|E| = 92$ } для G*

$\delta(G) = \min \deg(G) = 1$

$\Delta(G) = \max \deg(G) = 10$

$\mathrm{rad}(G) = 4$

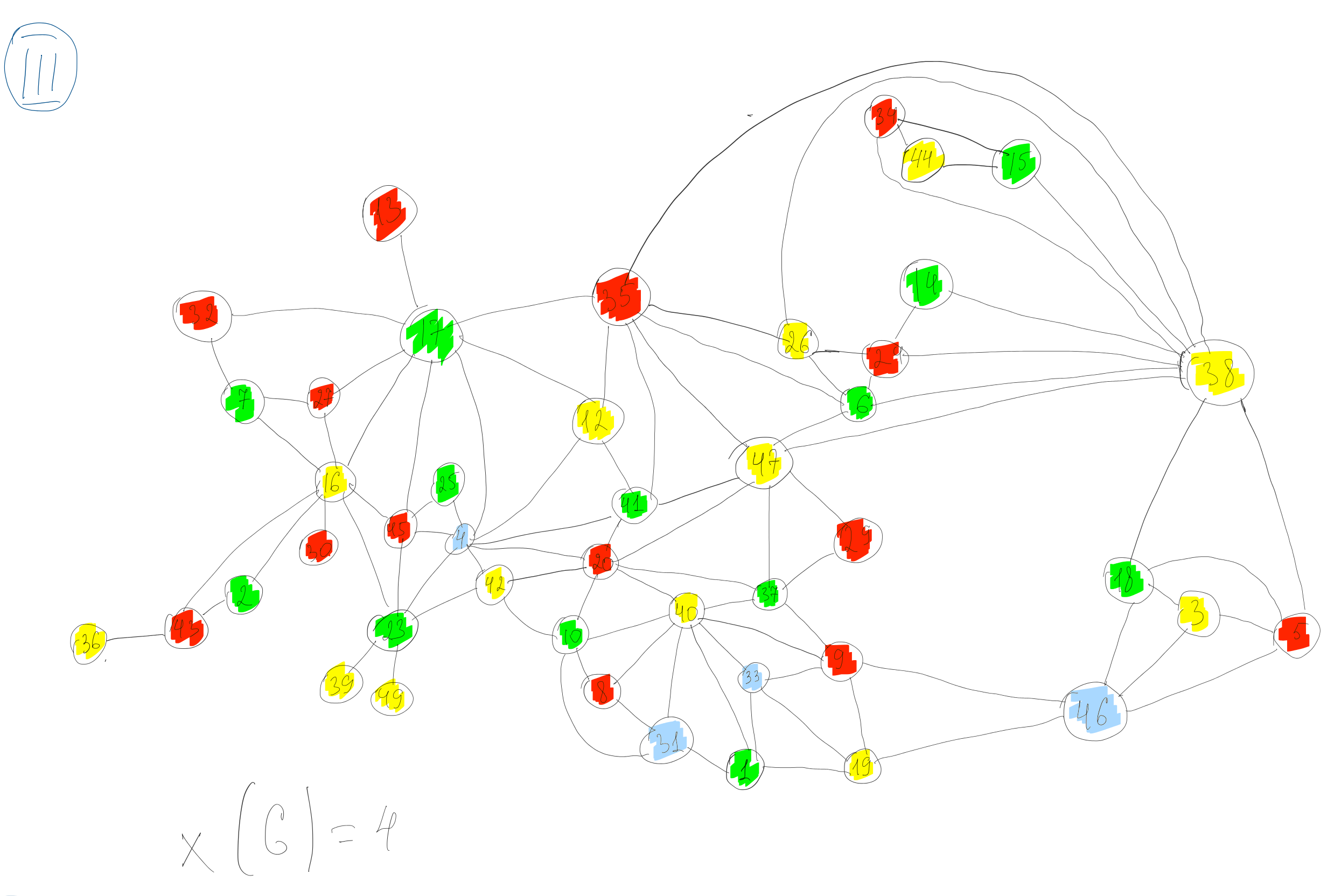
$\mathrm{diam}(G) = 8$

$\mathrm{center}(G) = 35$

} программа для подсчета – внизу, после работы

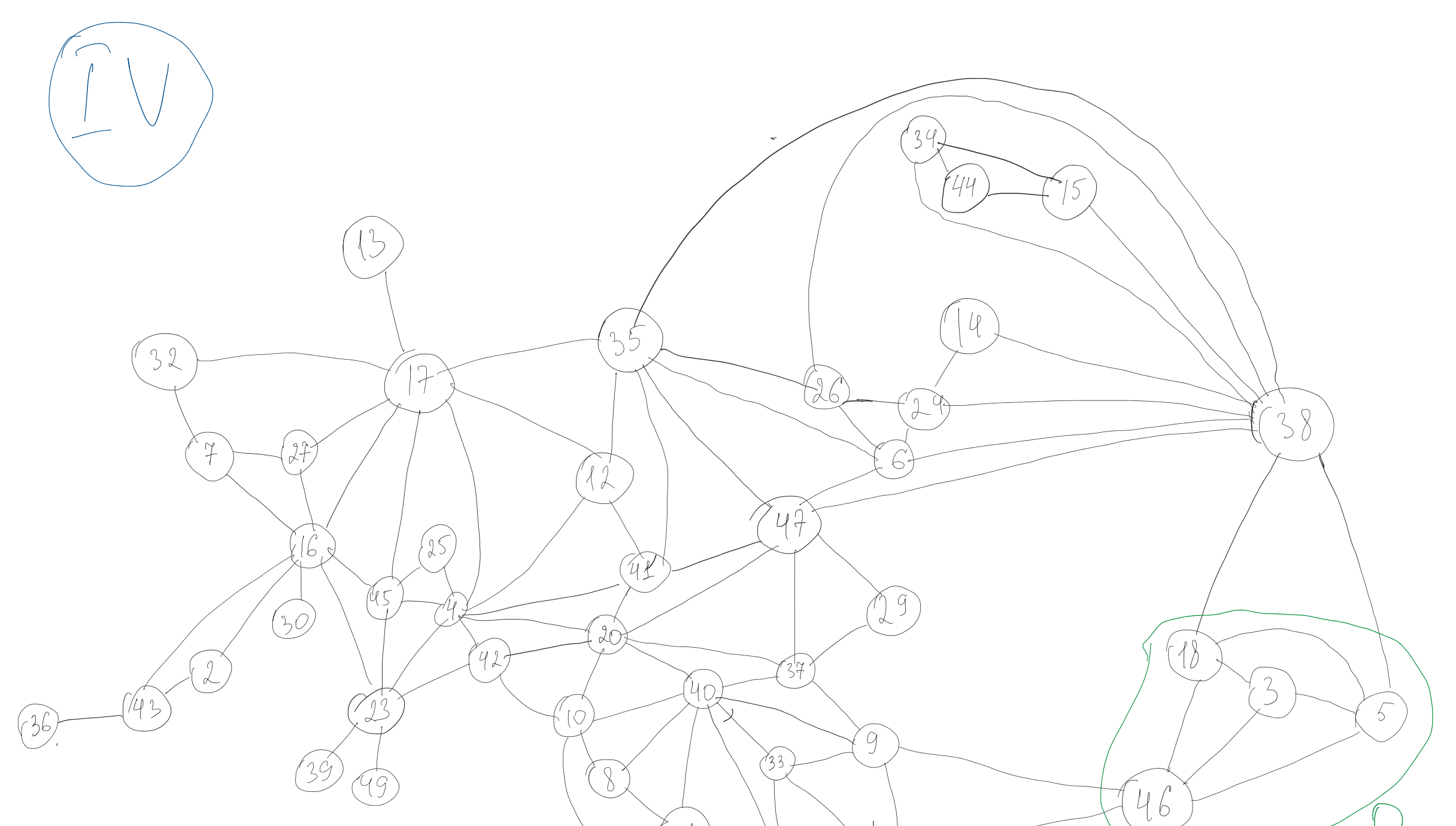
$z = |E| - |V| + k = 91 - 44 + 1 = 48$ (цикломатическое число) (для G)

## III



$\chi(G) = 4$

## IV



## V

Простой цикл W, содержащий все вершины графа G – есть Гамильтонов цикл

Так как в графе есть висячие вершины, например, 36, то Гамильтонова цикла быть не может, потому что для выхода из этой вершины нужно пройти по вершине, из которой ребро ведёт в висячую, второй раз => простого цикла W НЕТ

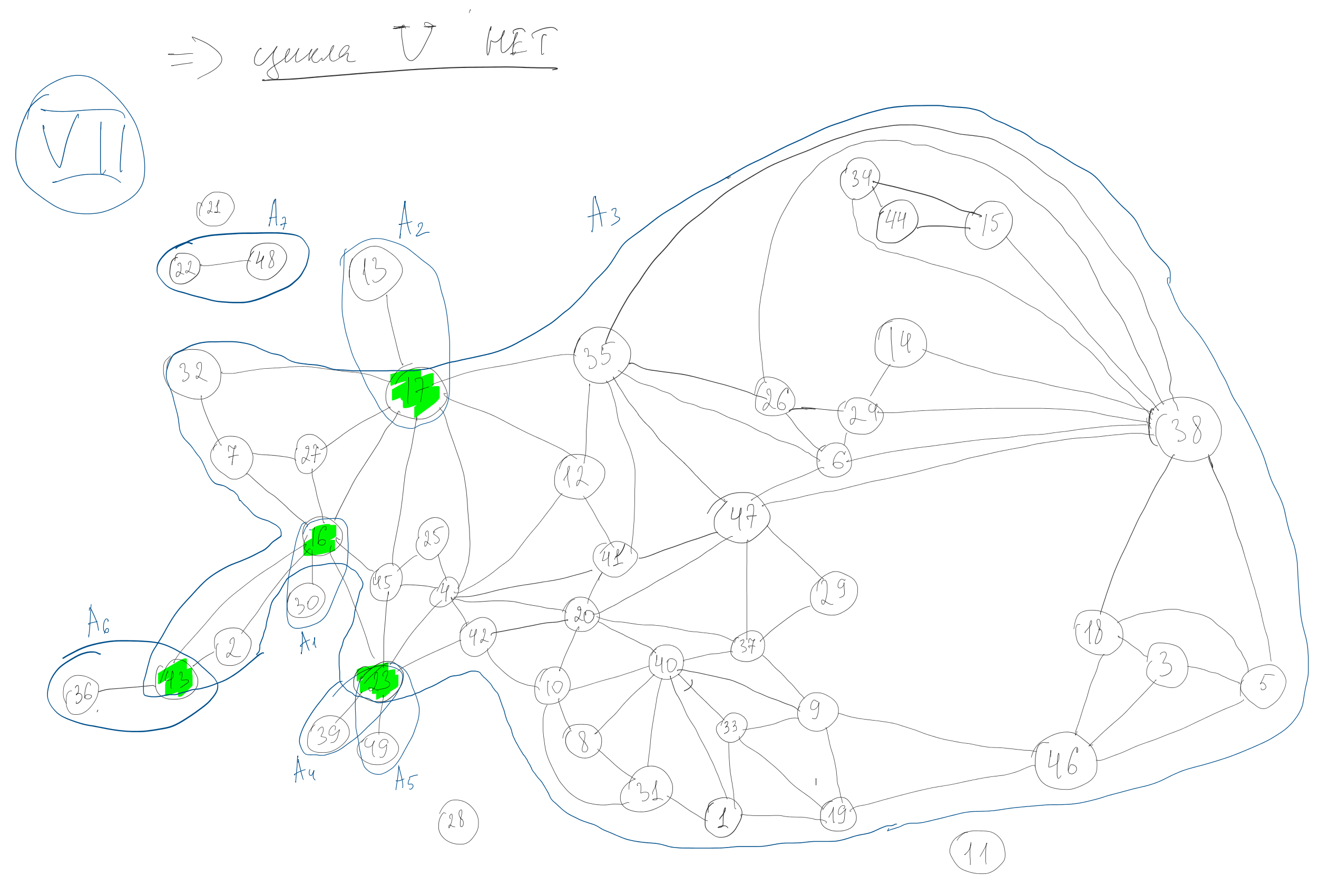
## VI

Цикл U, содержащий все ребра графа G – есть Эйлеров цикл.

Так как в графе есть вершины с нечётной степенью (3, 8, 46 и т.д.), то Эйлерова цикла в графе нет

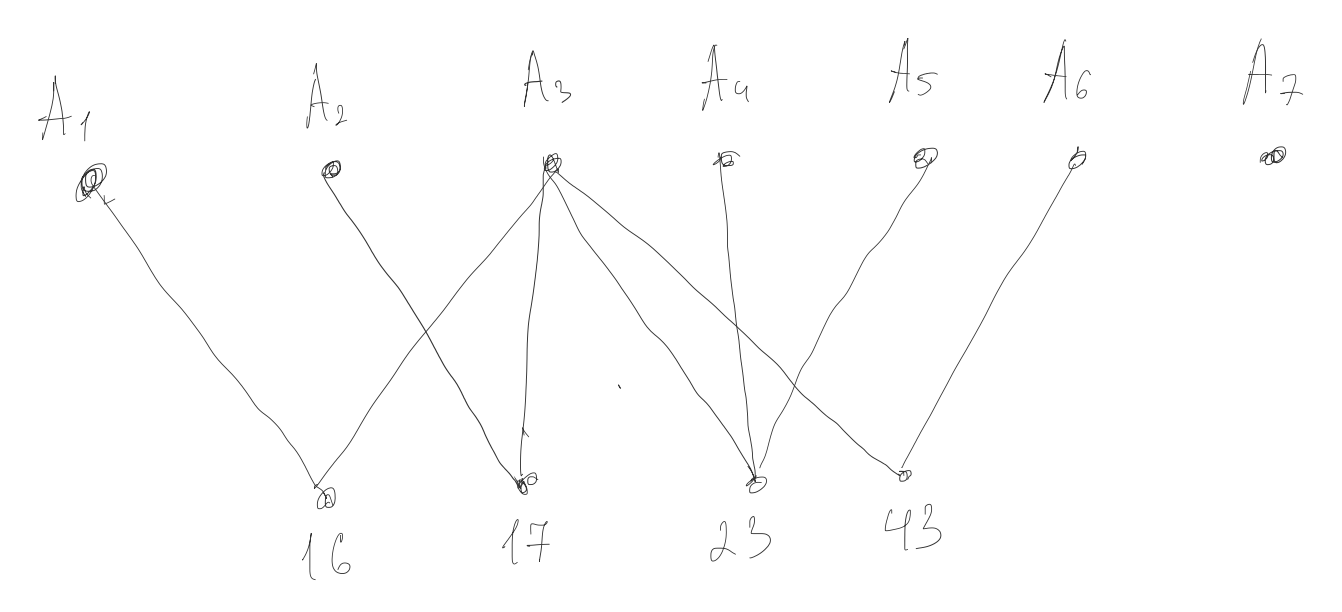
=> цикла U НЕТ

## VII

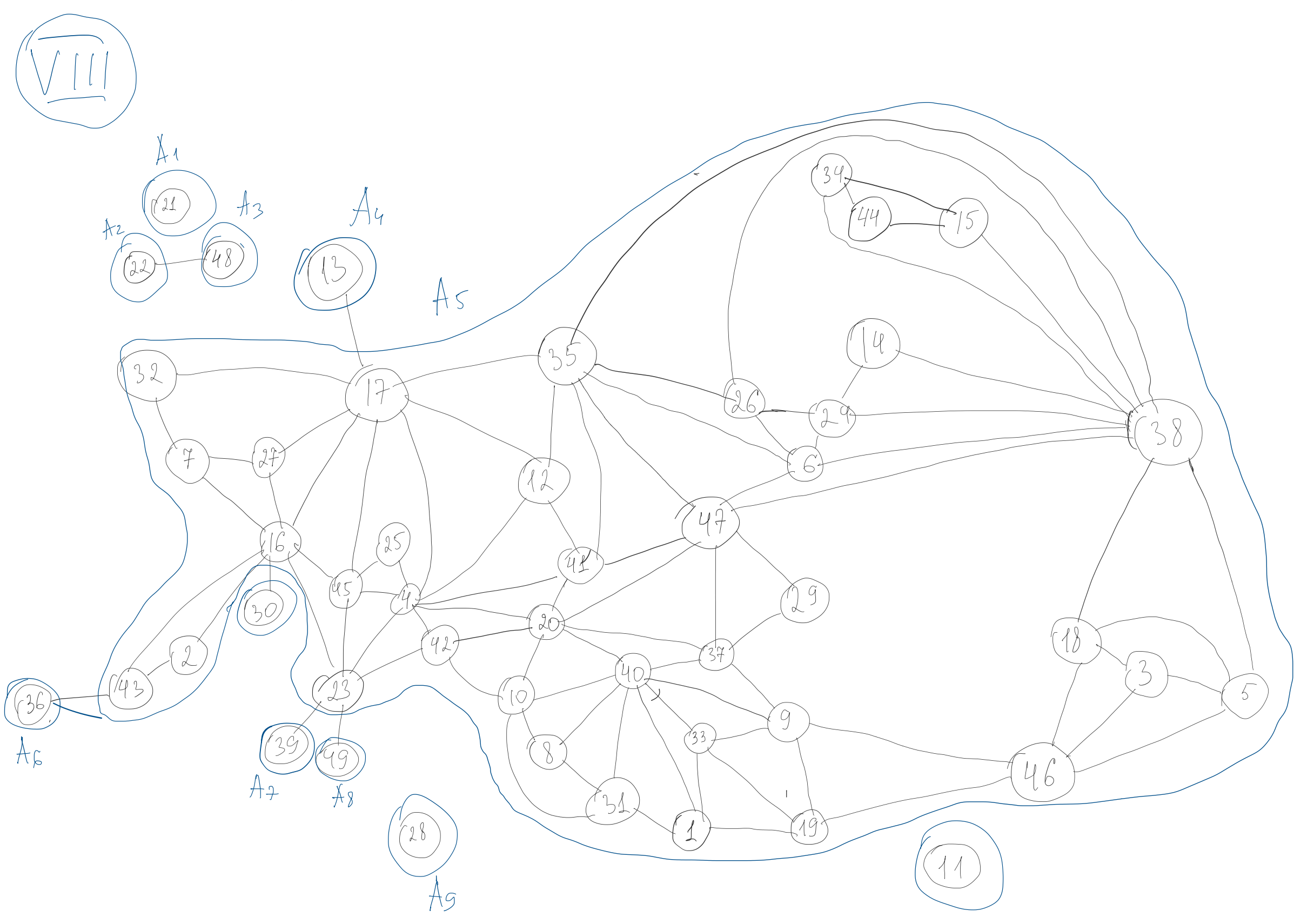


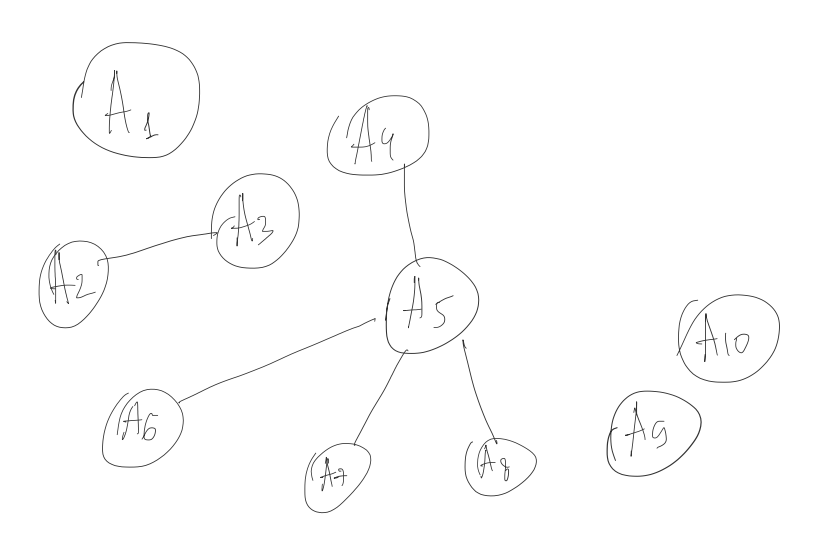
Блоки: $\{A_1, A_2, A_3, A_4, A_5, A_6, A_7\}$

Точки сочленения: $\{16, 17, 23, 43\}$

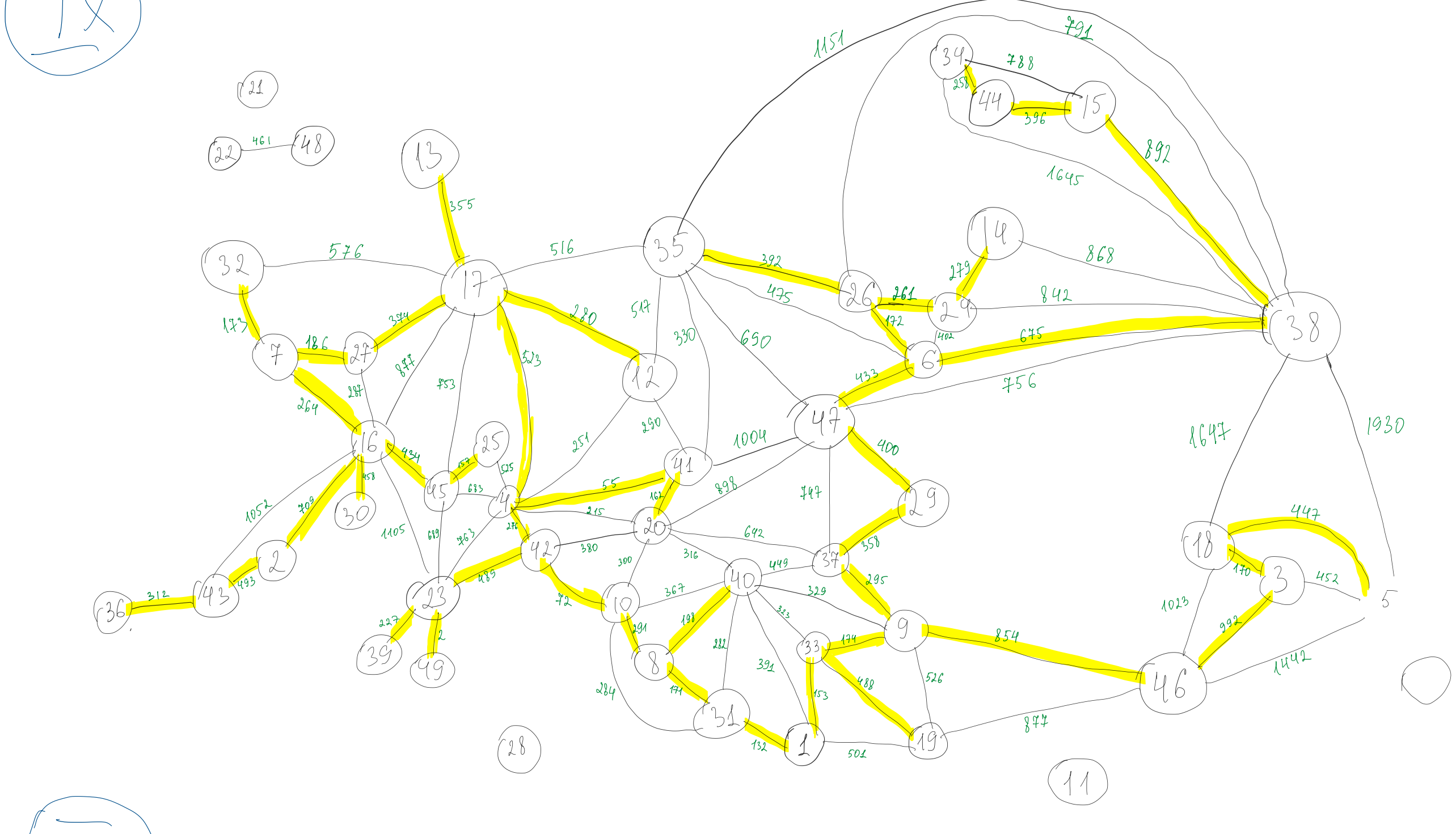


## VIII

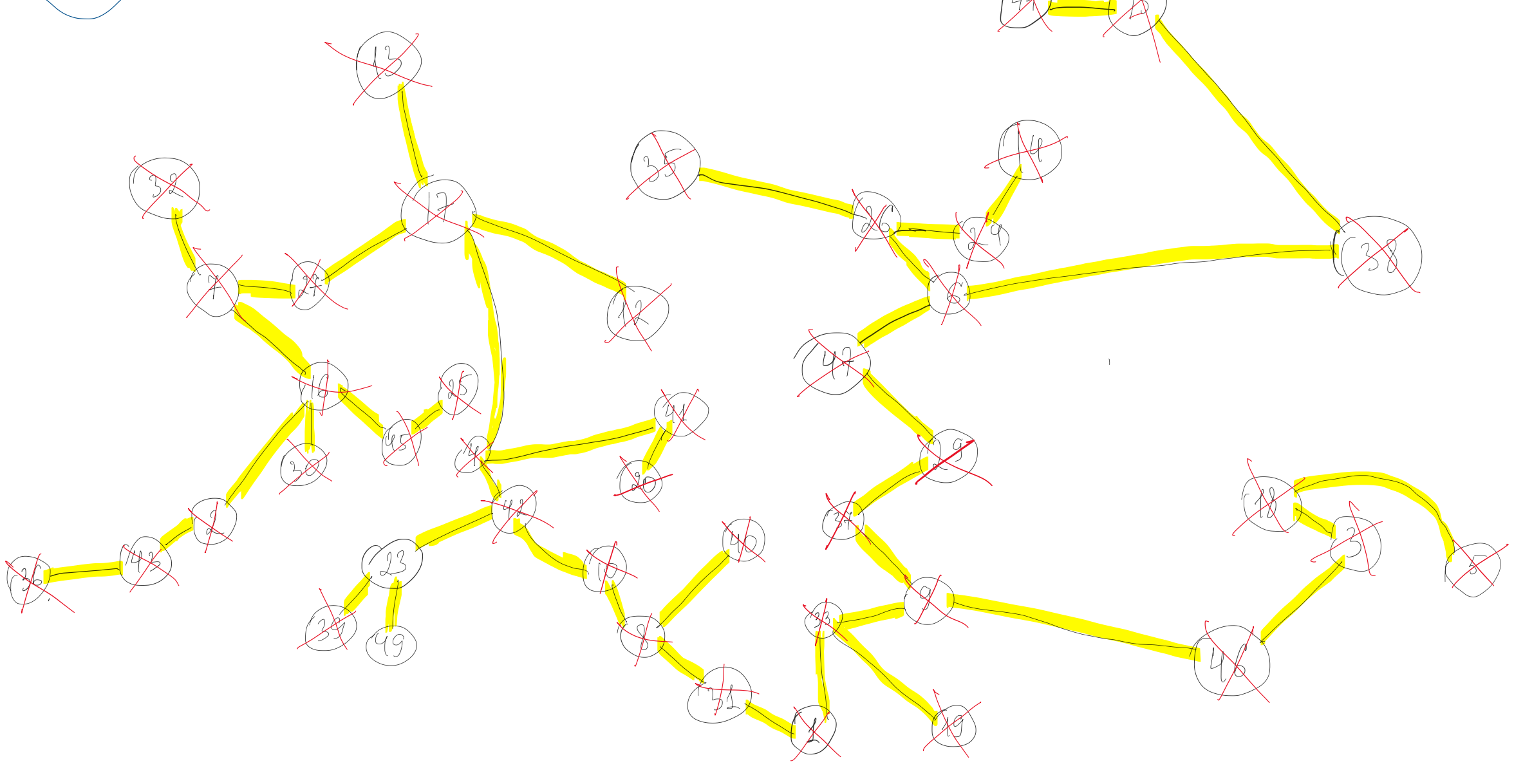




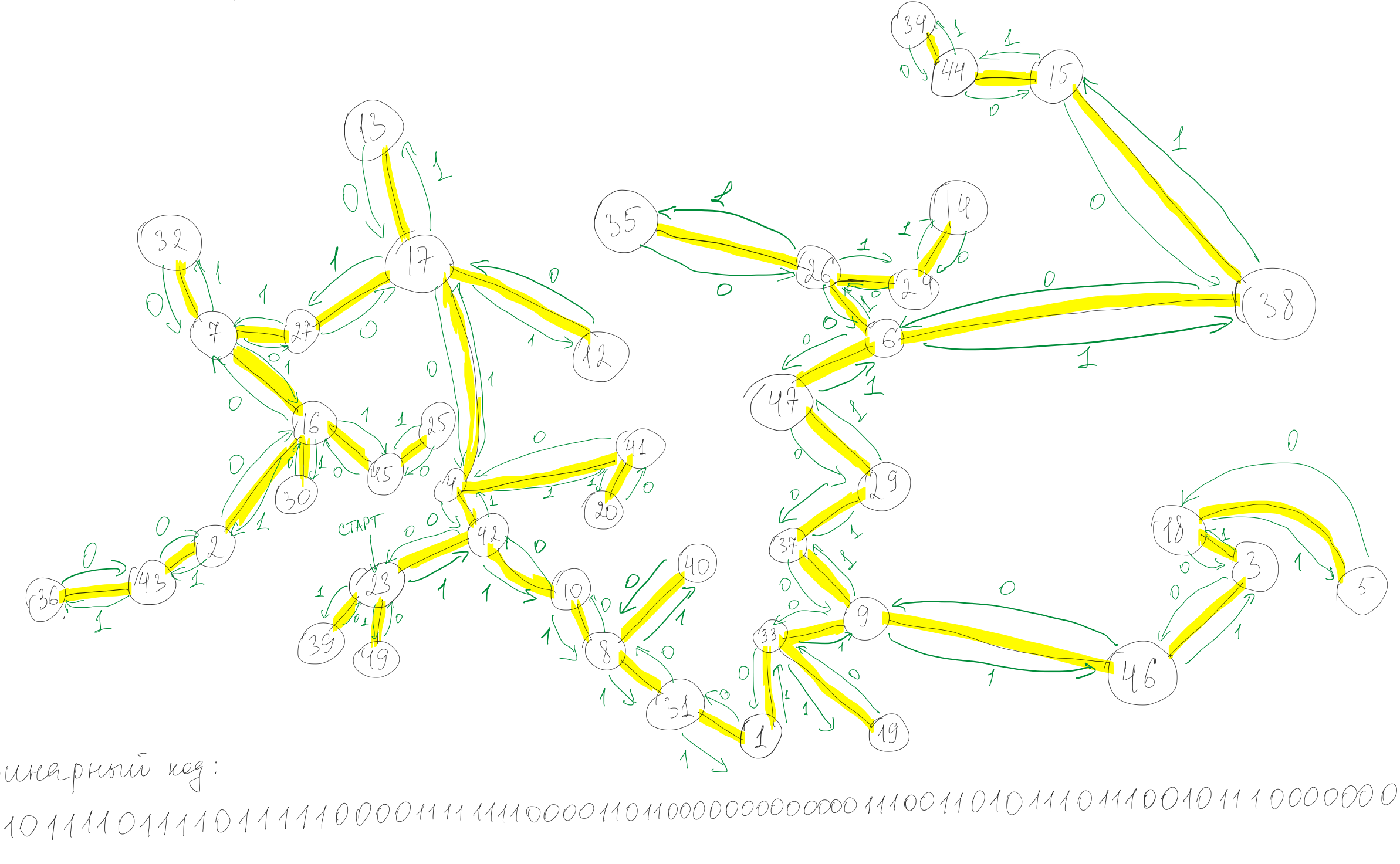
## IX



## X

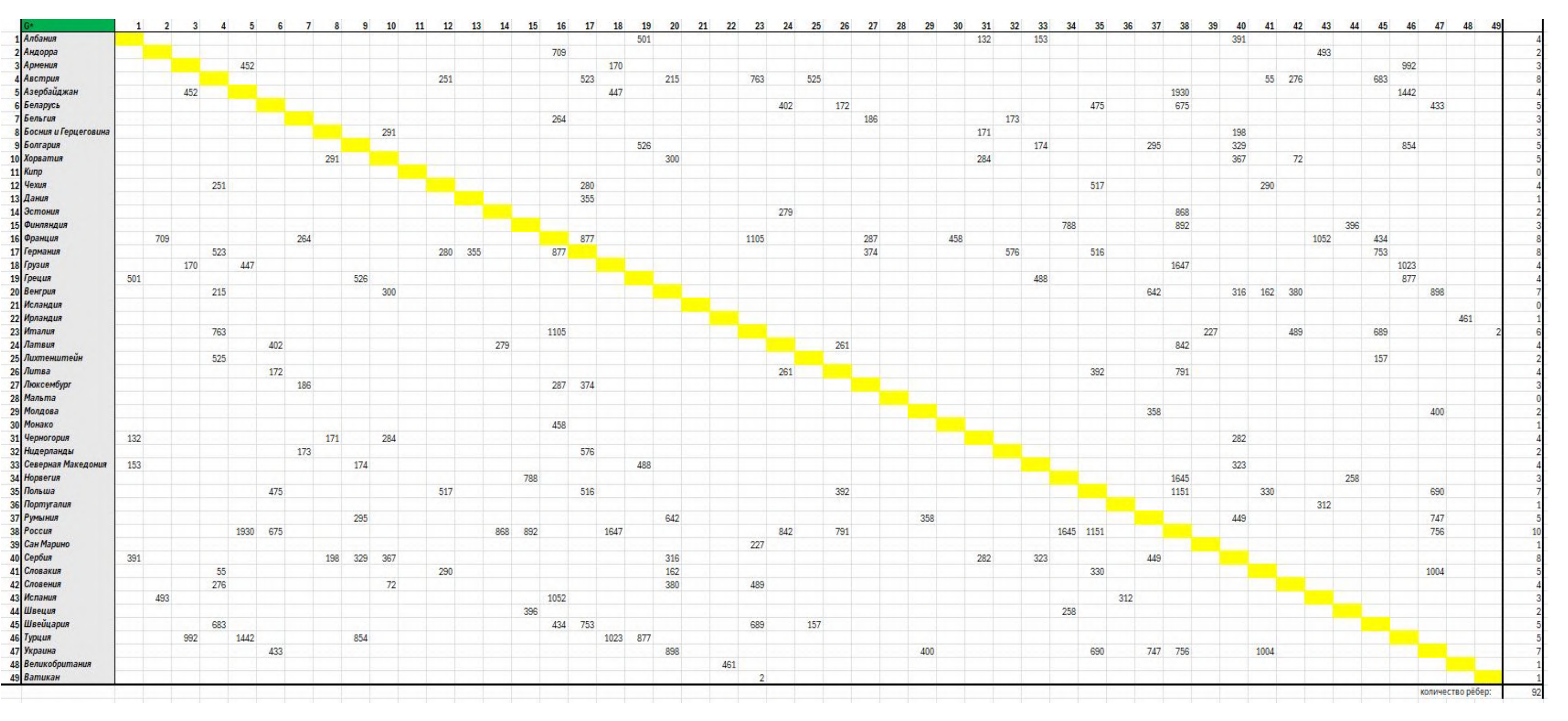


Код Прюфера: 18, 17, 17, 24, 3, 46, 33, 41, 26, 45, 16, 7, 44, 26, 6, 43, 23, 8, 4, 2, 16, 15, 38, 6, 47, 16, 7, 27, 17, 4, 42, 9, 29, 37, 9, 33, 1, 31, 8, 10, 42, 23

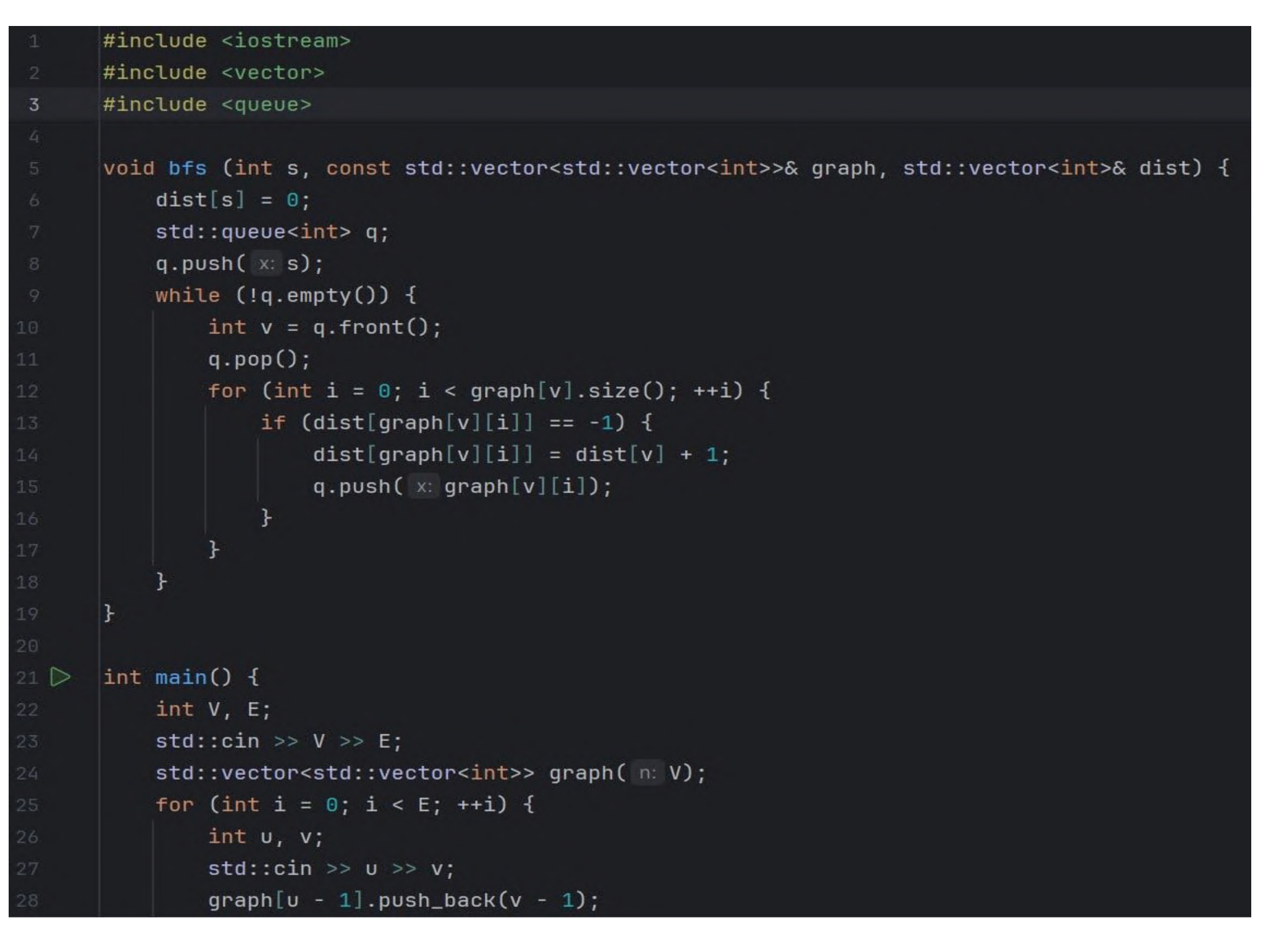


Бинарный код:

10101111011111011111100001111111100001101100000000000000011100110101110111001011110000000000



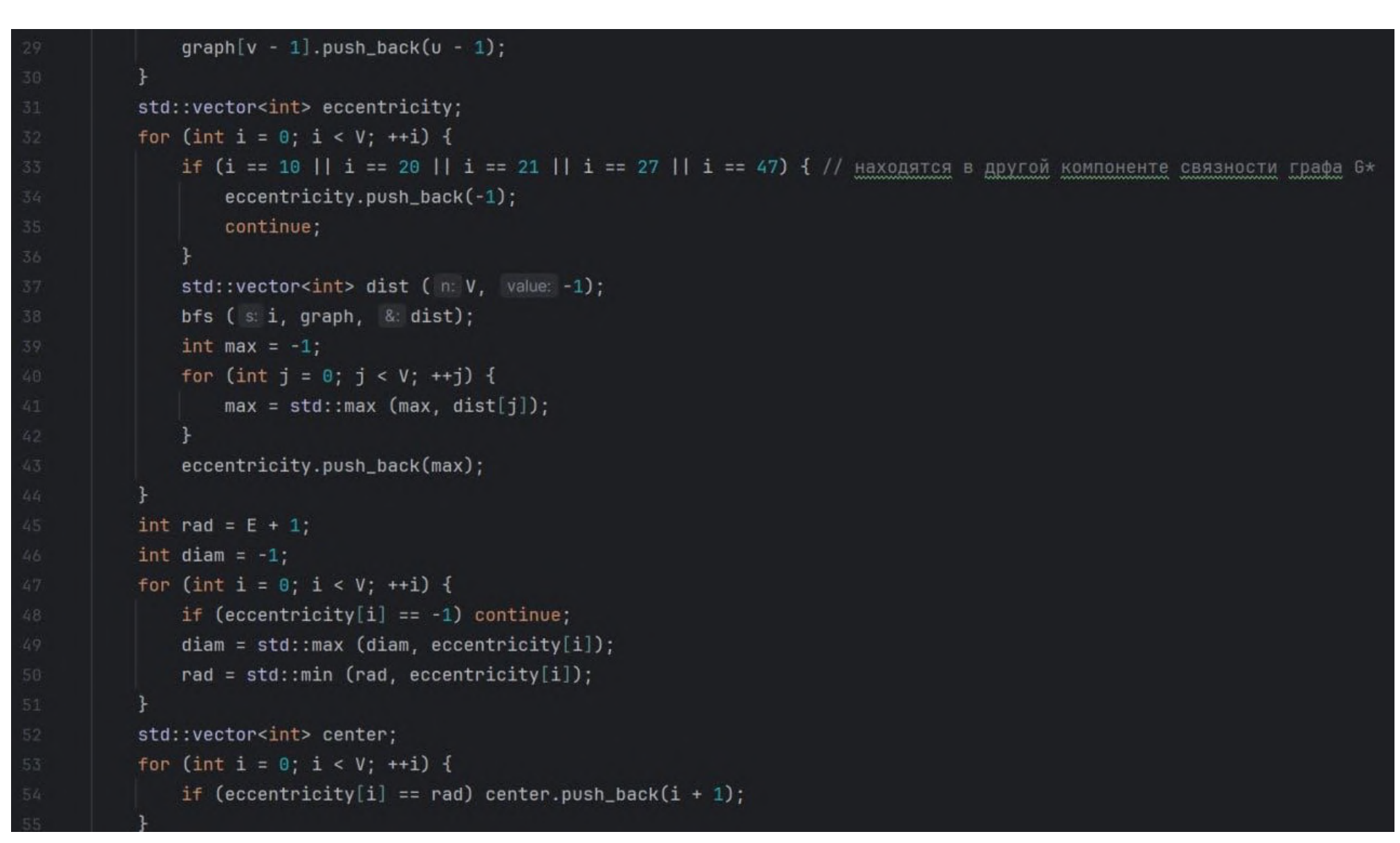
|  | G* | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 |  |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | Албания | | | | | | | | | | | | | | | | | | | 501 | | | | | | | | | | | | 132 | | 153 | | | | | | | 391 | | | | | | | | | | 4 |
| 2 | Андорра | | | | | | | | | | | | | | | | 709 | | | | | | | | | | | | | | | | | | | | | | | | | | | 493 | | | | | | | 2 |
| 3 | Армения | | | | | 452 | | | | | | | | | | | | | 170 | | | | | | | | | | | | | | | | | | | | | | | | | | | | 992 | | | | 3 |
| 4 | Австрия | | | | | | | | | | | | 251 | | | | | 523 | | | 215 | | | 763 | | 525 | | | | | | | | | | | | | | | | 55 | 276 | | | 683 | | | | | 8 |
| 5 | Азербайджан | | | 452 | | | | | | | | | | | | | | | 447 | | | | | | | | | | | | | | | | | | | | 1930 | | | | | | | | 1442 | | | | 4 |
| 6 | Беларусь | | | | | | | | | | | | | | | | | | | | | | | | 402 | | 172 | | | | | | | | | 475 | | | 675 | | | | | | | | | 433 | | | 5 |
| 7 | Бельгия | | | | | | | | | | | | | | | | 264 | | | | | | | | | | | 186 | | | | | 173 | | | | | | | | | | | | | | | | | | 3 |
| 8 | Босния и Герцеговина | | | | | | | | | | 291 | | | | | | | | | | | | | | | | | | | | | 171 | | | | | | | | | 198 | | | | | | | | | | 3 |
| 9 | Болгария | | | | | | | | | | | | | | | | | | | 526 | | | | | | | | | | | | | | 174 | | | | 295 | | | 329 | | | | | | 854 | | | | 5 |
| 10 | Хорватия | | | | | | | | 291 | | | | | | | | | | | | 300 | | | | | | | | | | | 284 | | | | | | | | | 367 | | 72 | | | | | | | | 5 |
| 11 | Кипр | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | 0 |
| 12 | Чехия | | | | 251 | | | | | | | | | | | | | 280 | | | | | | | | | | | | | | | | | | 517 | | | | | | 290 | | | | | | | | | 4 |
| 13 | Дания | | | | | | | | | | | | | | | | | 355 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | 1 |
| 14 | Эстония | | | | | | | | | | | | | | | | | | | | | | | | 279 | | | | | | | | | | | | | | 868 | | | | | | | | | | | | 2 |
| 15 | Финляндия | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | 788 | | | | 892 | | | | | | 396 | | | | | | 3 |
| 16 | Франция | | 709 | | | | | 264 | | | | | | | | | | 877 | | | | | | 1105 | | | | 287 | | | 458 | | | | | | | | | | | | | 1052 | | 434 | | | | | 8 |
| 17 | Германия | | | | 523 | | | | | | | | 280 | 355 | | | 877 | | | | | | | | | | | 374 | | | | | 576 | | | 516 | | | | | | | | | | 753 | | | | | 8 |
| 18 | Грузия | | | 170 | | 447 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | 1647 | | | | | | | | 1023 | | | | 4 |
| 19 | Греция | 501 | | | | | | | | 526 | | | | | | | | | | | | | | | | | | | | | | | | 488 | | | | | | | | | | | | | 877 | | | | 4 |
| 20 | Венгрия | | | | 215 | | | | | | 300 | | | | | | | | | | | | | | | | | | | | | | | | | | | 642 | | | 316 | 162 | 380 | | | | | 898 | | | 7 |
| 21 | Исландия | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | 0 |
| 22 | Ирландия | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | 461 | | 1 |
| 23 | Италия | | | | 763 | | | | | | | | | | | | 1105 | | | | | | | | | | | | | | | | | | | | | | | 227 | | | 489 | | | 689 | | | | 2 | 6 |
| 24 | Латвия | | | | | | 402 | | | | | | | | 279 | | | | | | | | | | | | 261 | | | | | | | | | | | | 842 | | | | | | | | | | | | 4 |
| 25 | Лихтенштейн | | | | 525 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | 157 | | | | | 2 |
| 26 | Литва | | | | | | 172 | | | | | | | | | | | | | | | | | | 261 | | | | | | | | | | | 392 | | | 791 | | | | | | | | | | | | 4 |
| 27 | Люксембург | | | | | | | 186 | | | | | | | | | 287 | 374 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | 3 |
| 28 | Мальта | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | 0 |
| 29 | Молдова | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | 358 | | | | | | | | | | 400 | | | 2 |
| 30 | Монако | | | | | | | | | | | | | | | | 458 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | 1 |
| 31 | Черногория | 132 | | | | | | | 171 | | 284 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | 282 | | | | | | | | | | 4 |
| 32 | Нидерланды | | | | | | | 173 | | | | | | | | | | 576 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | 2 |
| 33 | Северная Македония | 153 | | | | | | | | 174 | | | | | | | | | | 488 | | | | | | | | | | | | | | | | | | | | | 323 | | | | | | | | | | 4 |
| 34 | Норвегия | | | | | | | | | | | | | | | 788 | | | | | | | | | | | | | | | | | | | | | | | 1645 | | | | | | 258 | | | | | | 3 |
| 35 | Польша | | | | | | 475 | | | | | | 517 | | | | | 516 | | | | | | | | | 392 | | | | | | | | | | | | 1151 | | | 330 | | | | | | 690 | | | 7 |
| 36 | Португалия | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | 312 | | | | | | | 1 |
| 37 | Румыния | | | | | | | | | 295 | | | | | | | | | | | 642 | | | | | | | | | 358 | | | | | | | | | | | 449 | | | | | | | 747 | | | 5 |
| 38 | Россия | | | | | 1930 | 675 | | | | | | | | 868 | 892 | | | 1647 | | | | | | 842 | | 791 | | | | | | | | 1645 | 1151 | | | | | | | | | | | | 756 | | | 10 |
| 39 | Сан Марино | | | | | | | | | | | | | | | | | | | | | | | 227 | | | | | | | | | | | | | | | | | | | | | | | | | | | 1 |
| 40 | Сербия | 391 | | | | | | | 198 | 329 | 367 | | | | | | | | | | 316 | | | | | | | | | | | 282 | | 323 | | | | 449 | | | | | | | | | | | | | 8 |
| 41 | Словакия | | | | 55 | | | | | | | | 290 | | | | | | | | 162 | | | | | | | | | | | | | | | 330 | | | | | | | | | | | | 1004 | | | 5 |
| 42 | Словения | | | | 276 | | | | | | 72 | | | | | | | | | | 380 | | | 489 | | | | | | | | | | | | | | | | | | | | | | | | | | | 4 |
| 43 | Испания | | 493 | | | | | | | | | | | | | | 1052 | | | | | | | | | | | | | | | | | | | | 312 | | | | | | | | | | | | | | 3 |
| 44 | Швеция | | | | | | | | | | | | | | | 396 | | | | | | | | | | | | | | | | | | | 258 | | | | | | | | | | | | | | | | 2 |
| 45 | Швейцария | | | | 683 | | | | | | | | | | | | 434 | 753 | | | | | | 689 | | 157 | | | | | | | | | | | | | | | | | | | | | | | | | 5 |
| 46 | Турция | | | 992 | | 1442 | | | | 854 | | | | | | | | | 1023 | 877 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | 5 |
| 47 | Украина | | | | | | 433 | | | | | | | | | | | | | | 898 | | | | | | | | | 400 | | | | | | 690 | | 747 | 756 | | | 1004 | | | | | | | | | 7 |
| 48 | Великобритания | | | | | | | | | | | | | | | | | | | | | | 461 | | | | | | | | | | | | | | | | | | | | | | | | | | | | 1 |
| 49 | Ватикан | | | | | | | | | | | | | | | | | | | | | | | 2 | | | | | | | | | | | | | | | | | | | | | | | | | | | 1 |
|  |  | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | количество рёбер: | 92 |



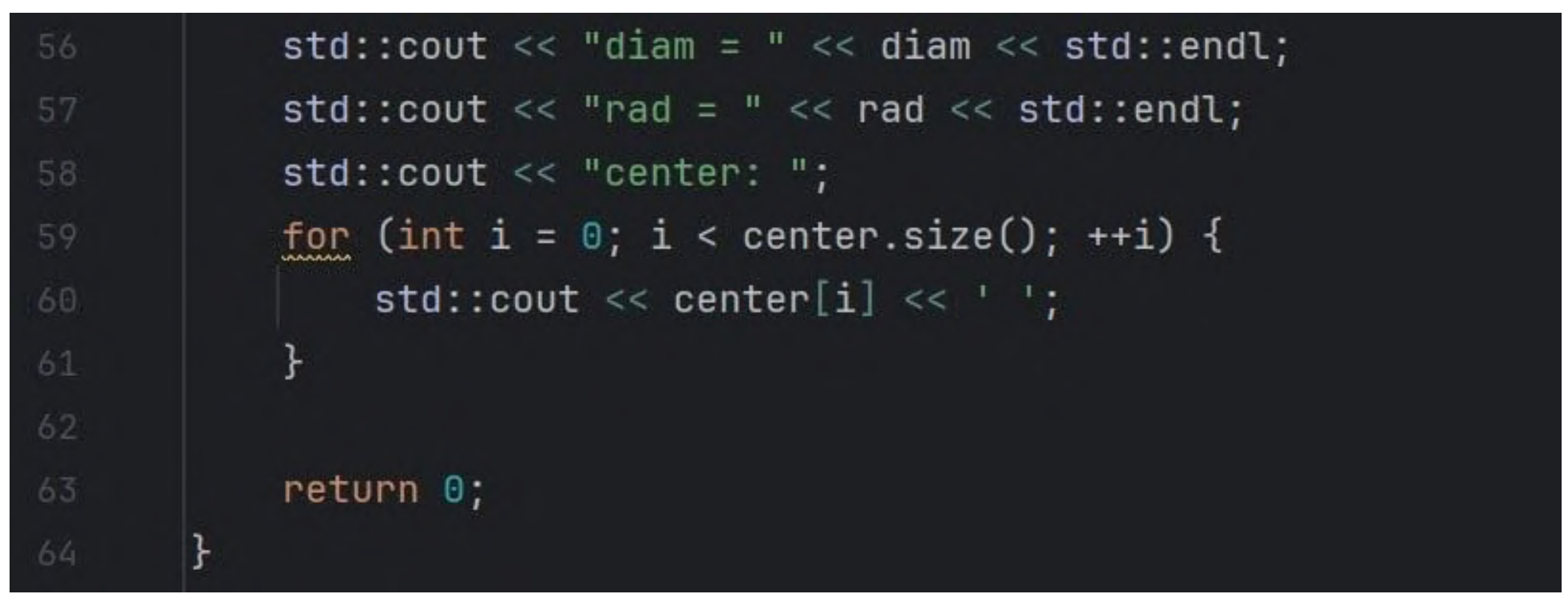
```
#include <iostream>
#include <vector>
#include <queue>

void bfs (int s, const std::vector<std::vector<int>>& graph, std::vector<int>& dist) {
    dist[s] = 0;
    std::queue<int> q;
    q.push( x: s);
    while (!q.empty()) {
        int v = q.front();
        q.pop();
        for (int i = 0; i < graph[v].size(); ++i) {
            if (dist[graph[v][i]] == -1) {
                dist[graph[v][i]] = dist[v] + 1;
                q.push( x: graph[v][i]);
            }
        }
    }
}

int main() {
    int V, E;
    std::cin >> V >> E;
    std::vector<std::vector<int>> graph( n: V);
    for (int i = 0; i < E; ++i) {
        int u, v;
        std::cin >> u >> v;
        graph[u - 1].push_back(v - 1);
```



```
        graph[v - 1].push_back(u - 1);
    }
    std::vector<int> eccentricity;
    for (int i = 0; i < V; ++i) {
        if (i == 10 || i == 20 || i == 21 || i == 27 || i == 47) { // находятся в другой компоненте связности графа G*
            eccentricity.push_back(-1);
            continue;
        }
        std::vector<int> dist ( n: V,  value: -1);
        bfs ( s: i, graph,  &: dist);
        int max = -1;
        for (int j = 0; j < V; ++j) {
            max = std::max (max, dist[j]);
        }
        eccentricity.push_back(max);
    }
    int rad = E + 1;
    int diam = -1;
    for (int i = 0; i < V; ++i) {
        if (eccentricity[i] == -1) continue;
        diam = std::max (diam, eccentricity[i]);
        rad = std::min (rad, eccentricity[i]);
    }
    std::vector<int> center;
    for (int i = 0; i < V; ++i) {
        if (eccentricity[i] == rad) center.push_back(i + 1);
    }
```



```
    std::cout << "diam = " << diam << std::endl;
    std::cout << "rad = " << rad << std::endl;
    std::cout << "center: ";
    for (int i = 0; i < center.size(); ++i) {
        std::cout << center[i] << ' ';
    }

    return 0;
}
```